MLH SOLANA HACKATHON · 2026

SkillChain

Learn Solana by doing. Every level is a real on-chain transaction, explained by an AI tutor.

skill-chain-lyart.vercel.app · Solana Devnet

THE PROBLEM

Blockchain is learned by reading. It should be learned by doing.

Jargon first

Wallets, keys, lamports and PDAs arrive as words, not experiences.

No real practice

Tutorials explain transactions. Beginners rarely send one.

Nothing to show

Course progress lives on someone else's server.

HOW IT WORKS

Three levels. Each one ends with a real transaction.

01

Wallet

Create your on-chain profile. Paying its rent proves you hold SOL and can sign.

02

Transactions

The program itself sends 0.01 SOL and completes the level.

03

Accounts & PDAs

Write a note the program stores in a PDA derived from your wallet.

Lesson → quiz → on-chain task → proof on Solana Explorer → points

BEYOND THE LEVELS

More than a course

AI tutor with follow-up questions

Gemini explains your real Devnet transaction, using verified on-chain facts only.

Speedrun duels

Your time comes from on-chain timestamps, so nobody can fake it.

Public, verifiable profile

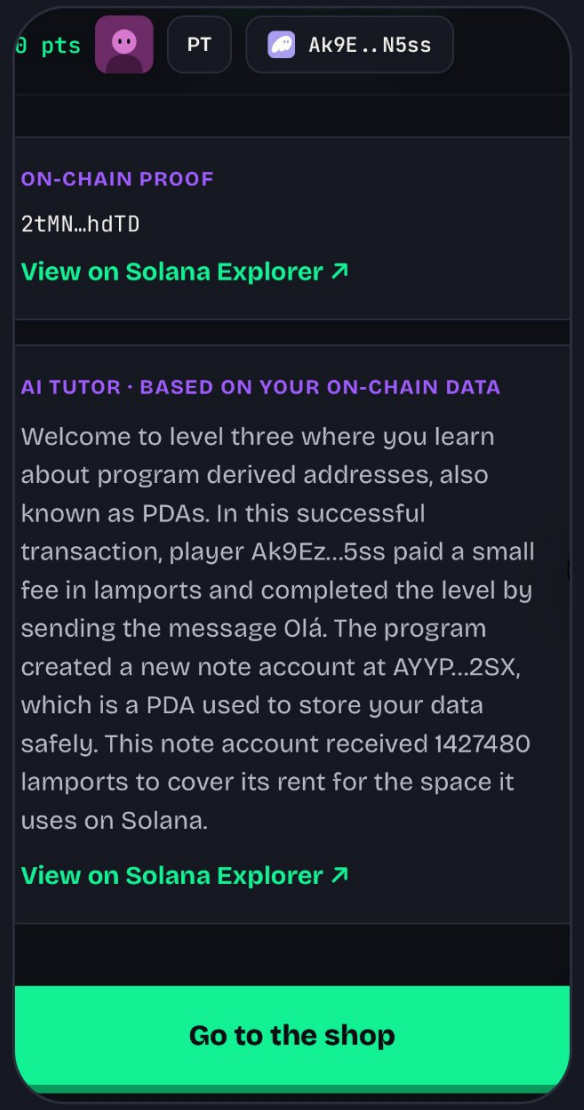
Anyone can check a player's level, items and note on Solana Explorer.

Shop and avatar

Points live in your profile PDA: earned, never bought or transferred.

Blink and demo mode

Accept a challenge in one click, or try everything without a wallet.



ARCHITECTURE

No backend database. No admin key.

Client

- Next.js and React on Vercel
- Phantom signs every transaction
- Hand-written client over web3.js
- English and Portuguese, mobile-first

Solana Devnet

- Anchor program in Rust
- Profile PDA: level, points, items
- Note PDA: message and timestamp
- Level order and maths enforced on-chain

Server routes

- /api/tutor: Devnet facts → Gemini
- /api/actions: Blink transaction
- Read the chain, never sign
- Gemini key stays on the server

WHY SOLANA

The chain is the referee.

Proof

Every level is a transaction anyone can check.

Ownership

Progress lives in the player's own PDAs, not on our server.

Speed and cost

Fast confirmations and tiny fees make learning by doing practical.

WHAT'S NEXT

From a weekend game to verifiable learning.

- More paths: tokens, NFTs, DeFi basics
- Non-transferable on-chain certificates
- Seasons and tournaments
- Classroom mode for universities and bootcamps

Try it
skill-chain-lyart.vercel.app

Code
github.com/afonsoPanzo/SkillChain

Program (Devnet)
AqdB1HaXB8wnvwGFTNsxTRrpSNMMVs2x5t4bz6UxU2X7